Optical and Quantum Electronics (2026) 58:175
https://doi.org/10.1007/s11082-026-08756-w

Tailoring the properties of Er-doped IZO thin films via controlled RF magnetron co-sputtering for photovoltaic applications

Mohsin Khan[1,2] · Qasim Ali[1] · Rashid Ali[3] · Muhammad Bilal Asif[4] · Ghazi Aman Nowsherwan[1] · Mahrukh Shafeeq[1] · Saira Riaz[1] · Mansun Chan[2] · Shahzad Naseem[1]

Received: 21 December 2025 / Accepted: 3 March 2026 / Published online: 17 March 2026
© The Author(s) 2026

Abstract

Er-doped indium zinc oxide (Er–IZO) thin films were deposited on glass substrates via radio-frequency magnetron co-sputtering, maintaining 90 W RF power on the IZO target while varying the Er DC power at 20, 30, and 40 W. XRD analysis showed that moderate Er incorporation (30 W) produced the largest crystallite size (31.76 nm), whereas excessive doping (40 W) reduced the coherence length to 28.40 nm due to increased lattice distortion. UV-VIS spectroscopy confirmed > 80% visible transparency for all samples, with the optical bandgap increasing from 3.28 eV (pristine) to 3.43 eV (20 W) and 3.80 eV (30 W) via the Burstein–Moss effect. A slight bandgap reduction to 3.50 eV at 40 W is attributed to defect-induced carrier compensation and enhanced scattering, which also caused minor transmission losses. Photoelectron spectroscopy revealed a non-monotonic work-function evolution: an initial decrease from 4.18 to 3.92 eV at 20 W, a modest increase to 3.94 eV at 30 W, and a larger rise to 4.32 eV at 40 W, driven by dopant clustering, carrier trapping, and electronic-structure modification. These results highlight the delicate balance between dopant incorporation, lattice integrity, and electronic behavior that governs the optimization of Er-doped IZO thin films as photovoltaic-relevant transparent conducting and optoelectronic materials.

Keywords IZO · Sputtering · TCO · Doping · Optoelectronic · Photovoltaic

✉ Mohsin Khan
mohsin.cssp@gmail.com; mkhanab@connect.ust.hk

[1] Centre of Excellence in Solid State Physics, University of the Punjab, Lahore 54590, Pakistan

[2] Department of Electronic & Computer Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong, SAR, China

[3] Department of Materials Science and Engineering, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi 23640, Pakistan

[4] Department of Mechanical and Aerospace Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong, SAR, China

1 Introduction

Transparent Conducting Oxides (TCOs) thin films have numerous diverse applications, e.g. solar cells, flat panel displays, flexible electronics, where the TCOs are applied as transparent electrodes and antireflection layers. TCOs must have working the principal necessity of good optical transmittance in the visible spectrum and the low electrical resistivity. Conventional TCOs, such as indium tin oxide (ITO), have been used; but newer ones that are flexible in their process and with better properties, such as IZO, are coming to the fore (Taylor et al. 2008, Zahid et al. 2022, Seyhan and Kartal 2023, Zeng et al. 2003).

IZO is a compositionally tunable transparent conducting oxide (TCO) that can be deposited in both amorphous and crystalline forms at relatively low processing temperatures. Amorphous IZO thin films exhibit high optical transparency (exceeding 90% in the visible region), good electrical conductivity (~3000 S/cm), and excellent thermal stability up to 600 °C (Taylor et al. 2008, Qiu et al. 2023, Park et al. 2021). The electrical and optical properties of IZO thin films have been extensively investigated, establishing their suitability as transparent conducting layers in optoelectronic and photovoltaic-relevant applications. The particular interest in IZO arises from its combination of high transparency and conductivity, which makes it a promising alternative to conventional TCOs. Murugesan et al. reported a comparative study of Al-doped and In-doped ZnO, highlighting the favorable optoelectronic characteristics of IZO for solar-related electrode applications (Murugesan et al. 2018). Furthermore, Christiansen et al. demonstrated that the incorporation of rare-earth dopants, such as erbium, can modify optical conversion processes in oxide materials, suggesting potential benefits for photovoltaic and photonic systems (Christiansen et al. 2020). To fully exploit these properties, advanced deposition techniques are required. In particular, RF magnetron sputtering enables the fabrication of highly conductive IZO films with reduced structural defects, which is essential for achieving efficient charge transport in transparent electrode applications (Koida et al. 2018, Li et al. 2022).

Oxide semiconductors incorporating erbium exploit the characteristic electronic transitions of Er^{3+} ions, most notably the $^4I_{13/2} \rightarrow {}^4I_{15/2}$ transition at approximately 1534 nm in the near-infrared region, which overlaps with the absorption range of crystalline silicon. Such down-conversion processes have been proposed as a potential route to convert sub-bandgap photons into higher-energy emissions, motivating interest in erbium-doped materials for photovoltaic-related and photonic applications. This optical activity makes erbium-doped oxides particularly attractive for systems requiring the coexistence of electrical conductivity and optical functionality (Kumi Barimah et al. 2020). In ZnO-based systems, erbium doping has been reported to modify defect-related emissions and enhance blue luminescence, effects commonly attributed to localized donor states and altered oxygen-vacancy populations. These changes arise from the introduction of shallow donor levels associated with Er incorporation, which influence carrier dynamics and optical absorption behavior (Ibiyemi et al. 2025). The incorporation of erbium into oxide lattices typically occurs through substitutional mechanisms, where Er^{3+} ions replace host cations. In zinc oxide, erbium can substitute Zn^{2+} without forming secondary phases when the dopant concentration remains below the solubility limit. X-ray diffraction studies have shown that moderate erbium incorporation preserves the host crystal structure while inducing subtle structural modifications and enhanced optical activity (Makir et al. 2021). In IZO systems, erbium ions introduce distinct luminescent characteristics, particularly near-infrared emissions that are of inter-

est for photovoltaic-relevant spectral engineering and optoelectronic applications (Wang et al. 2024). Beyond their optical response, erbium dopants have also been reported to influence film microstructure and defect distributions, contributing to improved uniformity and modified electronic behavior in oxide thin films (Bouabdalli et al. 2024). Recent advances in co-sputtering-based synthesis and characterization of IZO thin films provide a versatile platform for systematically tuning composition and dopant concentration. Fine control over erbium incorporation, combined with low-temperature deposition, offers opportunities to optimize transparent conducting oxide layers with properties relevant to flexible optoelectronic and photovoltaic electrode technologies (Lee et al. 2022, Kuwano et al. 2019, Tseng et al. 2020, Li et al. 2024).

The uniqueness of this study lies in the systematic investigation of how varying erbium sputtering power influences the structural, optical, and electronic properties of Er-doped IZO thin films, a parameter space that has not been comprehensively explored in previous reports. By correlating deposition conditions with key materials characteristics through an integrated set of structural, morphological, optical, and electrical analyses, this work provides a detailed understanding of dopant–lattice–carrier interactions in Er–IZO systems. Such insights offer practical guidance for identifying optimal dopant concentrations in transparent conducting oxide layers with relevance to photovoltaic and optoelectronic interfaces. In addition, the examination of erbium-induced photonic responses highlights the multifunctional nature of Er–IZO thin films, supporting their potential use in advanced optoelectronic and photovoltaic-relevant applications. Overall, these findings contribute to the broader development of tunable transparent conducting oxides for next-generation optoelectronic technologies.

2 Methodology and characterization

2.1 Experimental procedure

In this study, Er–IZO thin films were deposited on glass substrates by co-sputtering using an Ollital multi-target RF–DC magnetron system, as illustrated schematically in Fig. 1. High-purity targets were used, including indium zinc oxide (In_2O_3 and ZnO in a 9:1 weight ratio, 99.999% purity) and erbium (99.995% purity), each with a diameter of 50 mm. Argon was employed as the sputtering gas at a constant flow rate of 70 SCCM. Prior to deposition, the substrates were cleaned in acetone for 20 min and treated with Ar plasma for 5 min under a 400 V bias. The IZO and Er targets were pre-sputtered with the shutters closed to remove surface contaminants. The chamber was evacuated to a base pressure of 3.2 mPa, and each deposition was carried out at a working pressure of 400 mPa. The target-to-substrate distance was kept at 60 mm, and substrate rotation was applied to ensure uniform film thickness during co-sputtering. For Er-doped IZO films, the IZO target was driven with 90 W RF power for 15 min, while the Er target was powered under varying RF levels (0, 20, 30, and 40 W) for 5 min to achieve different Er incorporation levels.

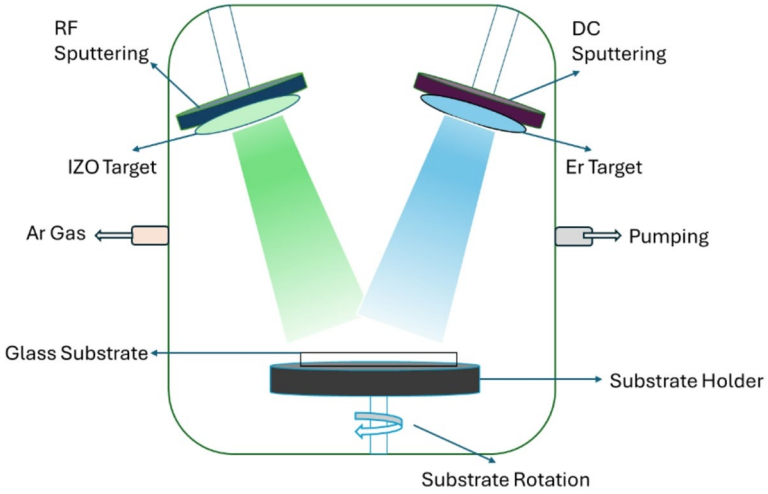


Fig. 1 Schematic illustration of co-sputtering

2.2 Characterization

The structure and stoichiometry of the synthesized thin films were characterized by X-ray diffraction (XRD) using a PANalytical X-ray diffractometer. Analysis was performed in a Bragg–Brentano (θ−2θ) mode with a monochromatic radiation source Cu-K alpha 1.54 A° with 40 kV accelerating voltage, and the tube current was 25 mA at room temperature. Atomic force microscopy (AFM) was used to investigate the surface morphology, surface roughness, and grain size of the samples in the tapping mode of operation using a Bruker AFM instrument. The SEM and elemental compositions of the samples were examined using energy-dispersive X-ray spectroscopy (EDX) in combination with a JEOL-6390 scanning electron microscope (SEM). The photoluminescence properties of the deposited thin films were characterized using an FS5 Spectrofluorometer. Thin-film transmittance and absorbance were measured using a Shimadzu 1800 spectrophotometer over a wavelength range of 200–1100 nm using ultraviolet-visible (UV-Vis) spectroscopy. Besides, a J.A. A Woollam VASE M-2000 ellipsometer was used to study optical characteristics. Electrical characterization was performed using Hall measurements, and the resistivity was measured using a four-point probe with the Hall-8800 Hall measurement system (Nanomat).

3 Results and discussion

3.1 X-ray diffraction

Figure 2 shows the XRD patterns of pure and Er-doped IZO thin films. All samples exhibit only the characteristic reflections of cubic In_2O_3 with a bixbyite structure, including the dominant (222) peak at ~30.6° (2θ) and the weaker (440) and (622) peaks at ~50.9° and ~60.6°, respectively. The absence of additional crystalline peaks indicates that the films

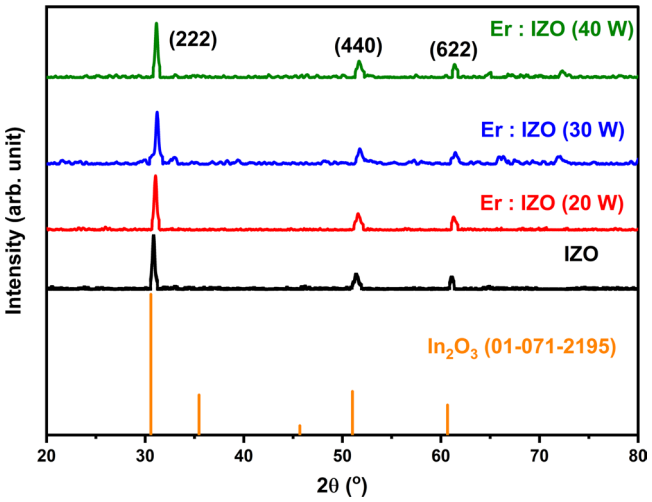


Fig. 2 XRD patterns of Pure and Er doped IZO thin films

remain phase-pure within the detection limits of the measurement across the investigated Er doping range.

With increasing Er sputtering power (20–40 W), the intensity of the (222) peak gradually decreases, indicating a reduction in long-range crystalline order and coherent domain size. This behavior is consistent with increased defect incorporation and lattice distortion associated with Er addition. A small but systematic shift of the (222) peak toward higher 2θ values is observed, amounting to approximately 0.11° between the undoped and 40 W films. According to Bragg's law, this corresponds to a slight reduction in interplanar spacing and lattice parameter.

Although Er^{3+} ions possess a slightly larger ionic radius (~0.89 Å) than In^{3+} (~0.80 Å), simple size-effect arguments alone do not govern the lattice response in this system. Moderate Er incorporation can reduce oxygen-vacancy concentration and promote film densification, leading to effective lattice contraction (Ágoston et al. 2009). In addition, compressive residual stress introduced during sputter deposition and local structural relaxation can further shift diffraction peaks toward higher 2θ values (Quinn et al. 2008). At higher Er power (40 W), defect accumulation, partial non-substitutional incorporation, or the presence of amorphous Er-rich regions may counteract vacancy compensation, resulting in the observed non-monotonic lattice behavior (Ali et al. 2026). Although the absolute magnitude of the peak shift is small, it is systematic across samples and exceeds instrumental resolution and fitting uncertainty, indicating a physically meaningful structural trend.

No crystalline Er or Er_2O_3 reflections are detected, suggesting that any Er-rich regions are either amorphous or below the XRD detection limit. The evolution of peak position and intensity is closely related to changes in crystallite size and microstrain, which were evaluated using the Scherrer formalism.

The crystallite size D, represents the average dimension of coherently diffracting crystalline domains within the thin film, and is commonly calculated from the full width at half maximum (FWHM) of the XRD peaks using the Scherrer equation:

$$D = \frac{K\lambda}{\beta\cos\theta} \tag{1}$$

where K = 0.9, λ is the X-ray wavelength, β is the FWHM (radians), and θ is the Bragg angle. As shown in Table 1, D increases from 23.20 nm (undoped) to a maximum of 31.76 nm at 30 W Er, before decreasing to 28.40 nm at 40 W. This indicates that moderate Er incorporation promotes crystallite growth, possibly through strain relaxation and enhanced adatom mobility, whereas excessive Er doping introduces lattice distortion that limits further coarsening.

$$\epsilon = \frac{\beta}{4\tan\theta} \tag{2}$$

The strain decreases from 4.42×10^{-3} (undoped) to -3.01×10^{-3} at 30 W Er, consistent with partial stress relaxation and oxygen-vacancy compensation. At 40 W Er, strain increases slightly to 3.34×10^{-3}, reflecting renewed lattice distortion at higher dopant levels.

The dislocation density ρ, is the length of dislocation lines per unit volume and is approximated by

$$\rho = \frac{1}{D^2} \tag{3}$$

follows a similar trend, decreasing with moderate Er doping and increasing again at 40 W. This suggests improved crystalline coherence at intermediate doping and enhanced defect formation at higher Er incorporation.

The lattice parameter a derived from the (222) reflection, decreases from 10.0376 Å (undoped) to 9.976 Å at 30 W Er, followed by a slight increase to 9.984 Å at 40 W. The unit-cell volume shows the same trend. This non-monotonic behavior reflects the competition between oxygen-vacancy compensation and densification (favoring contraction) and ionic radius mismatch and defect accumulation at higher Er concentrations (favoring slight re-expansion).

3.2 Atomic force microscopy (AFM) results

The topographical micrographs in Figs. 3(a-d) illustrate the topographical features of the different doping conditions. The residual pure IZO (image a) exhibits a comparatively smooth surface, moderate grain structure, and normal height (13 nm). As shown in Fig. 3b, the 20 W Er-doped sample exhibited better surface homogeneity, lower height variance

Table 1 XRD calculation parameters

parameters	IZO	Er: IZO (20 W)	Er: IZO (30 W)	Er: IZO (40 W)
Crystalline size (nm)	23.20	25.38	31.76	28.40
Microstrain (10^{-3})	4.4182	4.0406	3.0135	3.34007
Density Dislocation (nm^{-2})	1.8584	1.5520	0.9913	1.2397
a (A°)	10.0376	10.0047	9.976	9.984
V $(A°)^3$	1005.58	1001.41	997.29	997.52

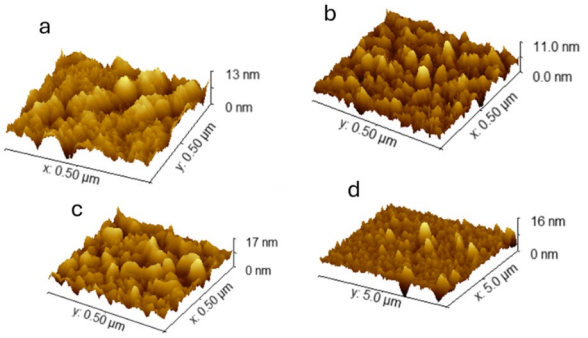


Fig. 3 AFM images of pure IZO and IZO films doped with Er at different DC sputtering powers: **a** Pure IZO, **b** 20 W, **c** 30 W, and **d** 40 W

Table 2 Roughness parameters calculated by AFM

Sample	Grain Size (nm)	Root mean square roughness R_{rms} (nm)	Average Roughness R_a (nm)
IZO	65	1.605	1.247
IZO: Er (20 W)	72	1.452	1.137
IZO: Er (30 W)	78	1.334	0.9654
IZO: Er (40 W)	51	2.246	1.777

(11 nm), and structured grain boundaries. The best quality is observed for the 30 W Er-doped film (Fig. 3c), which is characterized by the most uniform surface, regular grain structure, and regular height range (17 nm). The surface areas of the 40 W Er-doped sample (Fig. 3d) are also considerably degraded because of the increased surface roughness and decreased regularity of the grain structure, which proves the adverse outcomes of extreme Er-doping levels (Lee et al. 2025, Sambasivam et al. 2020). The correlated grain size measurements exhibited non-monotonic behavior with respect to the Er doping concentration. Pristine IZO has a grain size of 65 nm common to the sputtered IZO films under the room temperature environment (Table 2). When Er is doped at 20 W, the grain size is enhanced to 72 nm, indicating that the moderate incorporation of Er boosts grain growth by delivering more sources of nucleation and elevating the atomic mobility during film deposition (Cai et al. 2022).

At an Er doping level of 30 W, the grain size was at a maximum of 78 nm, which indicates the best scenario for crystalline growth. The improvement in grain size can be explained by the ability of Er^{3+} ions to promote grain growth through a decrease in the activation energy of grain boundary migration and encouragement of grain coalescence of minute grains. However, at the highest Er doping concentration (40 W), the grain size decreased to 51 nm (below the pristine IZO value). This decrease can be attributed to the overwhelming incorporation of Er, which causes structural destabilization, higher defect concentration, and the creation of Er-rich secondary phases that hamper the usual grain development mechanisms (Ren et al. 2019).

It is important to note that the grain size measured by AFM and the crystallite size calculated from XRD represent different physical quantities and should not be directly equated. The crystallite size obtained from XRD corresponds to the average dimension of coherently diffracting domains within the film, whereas the AFM grain size reflects surface morpho-

logical features that may consist of one or multiple crystallites. In polycrystalline oxide thin films, a single surface grain can contain several crystallites separated by low-angle grain boundaries. Therefore, AFM and XRD provide complementary information on surface morphology and internal crystalline coherence, respectively.

The measurements of the RMS roughness (R_{rms}) indicate a systematic relationship with the Er doping concentration, similar to the grain size variations. Clean IZO has an R_{rms} of 1.605 nm, which is consistent with those reported for sputtered IZO films(Pan et al. 2005). Doping to 20 W decreases the R_{rms} to 1.452 nm, demonstrating that there are better surfaces after doping because the atomic mobility is increased, and the densification of the films is more successful (Seyhan and Kartal 2023, Li et al. 2022). The lowest R_{rms} 1.334 nm is made when 30 W Er is doped, and hence is the best optimum doping situation that forms the smoothest surface morphology. Regarding the explanation of the same, such an effect is attributed to the Er atoms aiding surface diffusion and stimulating a homogenous growth of the film (Juraić et al. 2022). R_{rms} is also significantly affected by an increase of 2.246 nm at a 40 W Er doping level, which results in the deterioration of the surface at high doses of Er doping. This increase in roughness is probably the result of the cluster developed in the presence of a high density of defects and the heterogeneous development of the film because of the strain in the structure and the division of phases (Sambasivam et al. 2020). The average roughness (R_a) values exhibit the same trends as the R_{rms}, producing smaller absolute values. Doping IZO with Er at 20 W minimized (1.137 nm) R_a and finally reached a maximum (1.777 nm) at 40 W, having passed its lowest (0.9654 nm) at 30 W; it started at 1.247 nm in pristine IZO (Table 2).

3.3 Energy-dispersive X-ray (EDX) results

Energy-dispersive X-ray (EDX) spectroscopy was used to quantify the elemental composition of the Er-doped IZO thin films, and the results are presented in Fig. 4a–d. The undoped IZO film shows only the expected emission lines of In (L_α at ~3.3 keV), Zn (K_α at ~8.6 keV), and O (K_α at ~0.52 keV). The In: Zn ratio closely matches the target stoichiometry, while the O: (In + Zn) ratio is slightly below the ideal value of 1.5, consistent with the presence of oxygen vacancies that act as intrinsic free-carrier donors in IZO. Upon applying 20 W DC power to the Er target, a weak Er M_α peak appears at ~5.8 keV, corresponding to approximately 1.2 at% Er. This initial incorporation is accompanied by a slightly decreased Zn fraction and slight increase in the oxygen concentration. These changes indicate that Er3 + ions mainly replace In^{3+} ion (or sometimes located at interstitial site) and partially passivate pre-existing oxygen vacancies, consistent with the decreased lattice micro strain of XRD study.

By increasing the DC power, the Er content increases systematically, and for 30 W and 40 W, the content is approximately 2.3 and 3.5%, respectively. On increasing the Er concentration, the Zn fraction goes further down and the O: (In + Zn+Er) ratio gets closer to the ideal value of 1.5, indicating progressive oxygen-vacancy compensation by Er-O bonding. Importantly, no anomalous enrichment or segregation of oxygen is observed, and the In/Zn ratio remains within the expected compositional window, supporting the conclusion that Er is incorporated into the IZO lattice without forming secondary phases. The systematic increase in Er content, the controlled variation in cation ratios, and the stable oxygen stoichiometry collectively support the XRD findings: erbium is incorporated substitution-

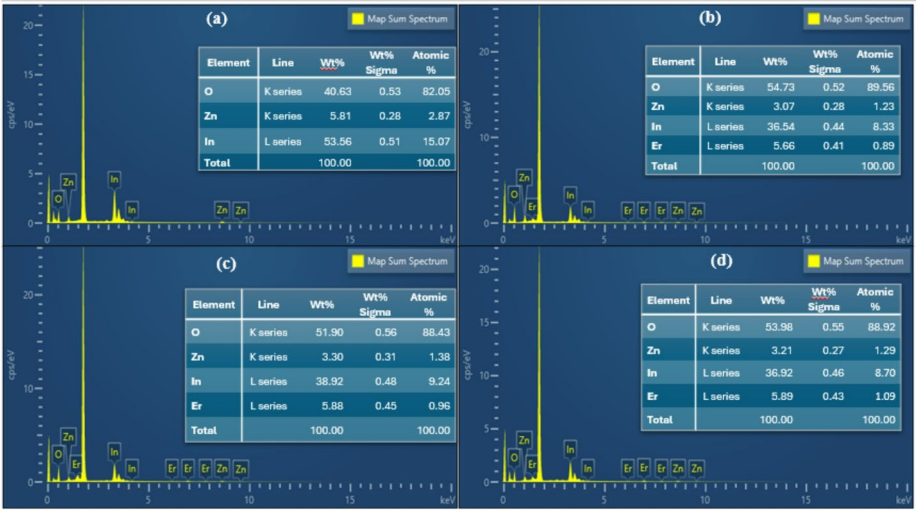


Fig. 4 EDX spectra with quantified compositions for (**a**) pure IZO and Er-doped IZO films at (**b**) 20 W, **c** 30 W, and **d** 40 W DC sputtering power

ally or interstitially within the IZO lattice, introducing only minor perturbations in lattice parameters and defect populations without causing phase separation or impurity formation.

3.4 X-ray fluorescence (XRF) analysis

X-ray fluorescence (XRF) spectroscopy was employed to obtain quantitative compositional information for the Er-doped IZO thin films, complementing the semi-quantitative elemental trends obtained from EDX (Fig. 5a-d). Unlike EDX which is surface sensitive and affected by matrix effects—XRF provides bulk-averaged, thickness-independent quantification, enabling reliable determination of oxide fractions and dopant loading in the films. This additional analysis is necessary because for co-sputtered multi-cation oxides there are often subtle changes of composition which mean that high-accuracy elemental ratios are necessary for the correct interpretation of the structural and optical behavior. The XRF spectra of the undoped and doped films show the typical emission lines of the constituents In ($L\alpha_1$, $L\beta_1$, $L\beta_2$), Zn ($K\alpha$, $K\beta_1$), Er ($L\alpha_1$, $L\beta_1$, $L\beta_2$, $L\gamma_1$ in doped samples), and Si ($K\alpha$ from the glass substrate). Peaks associated with the Rh excitation source (Rh $L\alpha_1$, $L\beta_1$, $L\beta_2$, $L\gamma_1$, $K\alpha$) are also detected together with lower intensity signals from Mg, K, Ca, Fe and Cu due to either substrate impurities or environmental background. In the undoped IZO film, Na, P and S elements are observed instead of Er, in line with residual substrate or glass elements. It should be noted that XRF measurements provide bulk elemental composition in mass percentage (wt%), whereas discussions of substitutional incorporation and defect chemistry refer to atomic-scale local environments inferred from structural and optical analyses.

Quantitative fitting was performed by using a two-layer model composed of Layer 1 (thin film) In_2O_3, ZnO and Er_2O_3 (for doped samples) and Layer 2 SiO_2 (fixed corresponding to the glass substrate). The goodness-of-fit coefficients were observed to vary from 0.0266 to 0.0445 and hence the spectral modeling was stable and convergent for all the samples. As

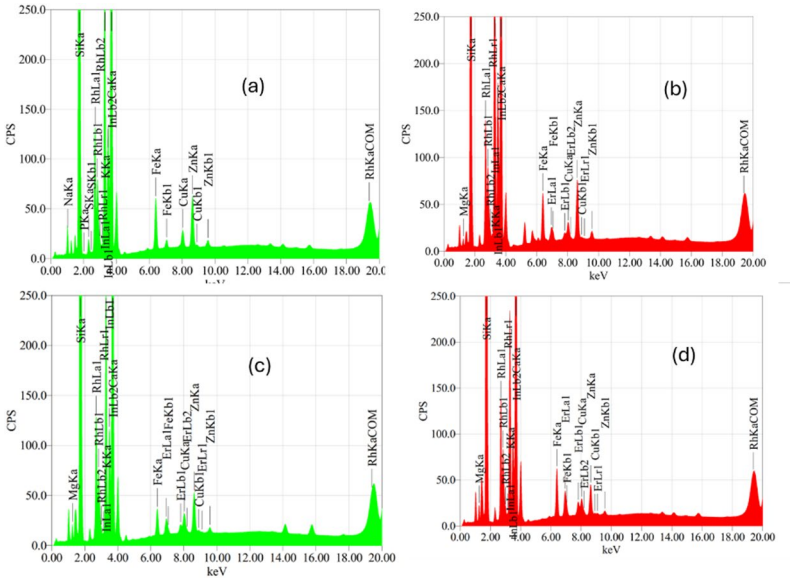


Fig. 5 XRF spectra of (**a**) pure IZO, **b** Er-doped IZO at 20 W, **c** Er-doped IZO at 30 W, and **d** Er-doped IZO at 40 W DC sputtering power

Table 3 XRF quantification of In_2O_3, ZnO, and Er_2O_3 mass percentages in pure IZO and Er-doped IZO films deposited at different DC sputtering powers

Sample	In_2O_3 (mass%)	ZnO (mass%)	Er_2O_3 (mass%)
IZO (undoped)	90.35	9.645	0
Er: IZO 20 W	88.06	8.356	3.589
Er: IZO 30 W	85.30	8.054	6.645
Er: IZO 40 W	79.96	7.344	12.690

the Er DC power increased from 20 W to 40 W, the Er_2O_3 content increased systematically from 3.589 mass% → 6.645 mass% → 12.690 mass%, confirming progressive enhancement in Er incorporation with higher sputtering power. Correspondingly, the In_2O_3 fraction decreased from 88.060 mass% → 79.960 mass%, and ZnO declined slightly from 8.356 mass% → 7.344 mass%. This inverse trend reflects compositional competition among cations during co-sputtering, where increased Er flux reduces the relative incorporation of In and Zn into the growing film (Table 3).

3.5 Photoluminescence spectroscopy

Photoluminescence spectra of Er-doped indium zinc oxide (IZO) thin films, shown in Figs. 6a–d, reveal multiple emission features that reflect the optical response of this rare-earth-doped oxide system. The systematic variation in Er sputtering power (20 W, 30 W, and 40 W) enables an examination of concentration-dependent luminescence behavior. The introduction of erbium into the IZO matrix modifies the emission characteristics of the host, giving rise to additional radiative features that are consistent with erbium-related optical activity reported in similar oxide systems. Erbium ions are generally expected to be present in the trivalent state (Er^{3+}, $4f^{11}$ electronic configuration), which introduces localized

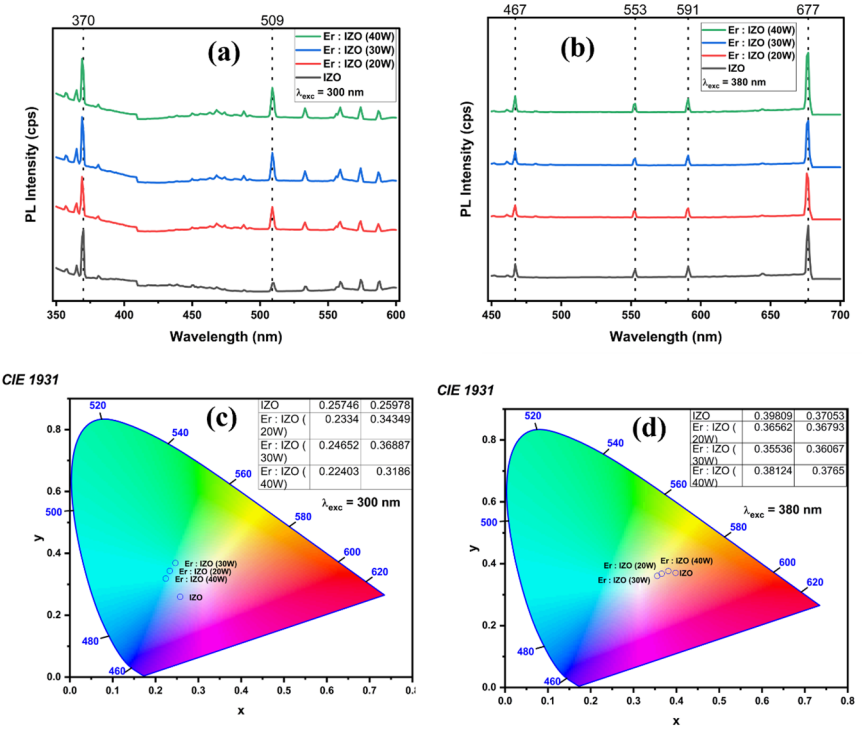


Fig. 6 PL spectra of pure and Er-doped IZO films excited at (**a**) 300 nm and (**b**) 380 nm, with corresponding CIE chromaticity plots shown in (**c**) and (**d**), respectively

energy levels within the bandgap of oxide hosts and can contribute to sharp emission bands observed in photoluminescence spectra (Sunde et al. 2014).While erbium incorporation is commonly attributed to substitutional or near-substitutional sites in oxide lattices, the present PL results should be regarded as indicative rather than definitive evidence of specific site occupancy, as direct chemical-state analysis was not performed.

Under 380 nm excitation (Fig. 6b), several discrete emission peaks are observed that are consistent with intra-4f transitions of Er^{3+} ions reported in the literature. In particular, the prominent emission at approximately 677 nm can be associated with the $^4F_{9/2} \rightarrow {}^4I_{15/2}$ transition, a characteristic red emission commonly observed in erbium-doped oxide materials. Additional emission features at approximately 467, 553, and 591 nm are consistent with transitions involving higher excited Er^{3+} energy levels, such as $^2H_{11/2}$, $^4S_{3/2}$, and $^4F_{7/2}$. The presence of multiple emission bands suggests that excitation at 380 nm effectively populates several excited states of Er^{3+} ions, either through direct optical excitation or via interaction with the host matrix, as widely reported for rare-earth-doped oxides (Kien et al. 2024, Liao et al. 2018).

In contrast, excitation at 300 nm (Fig. 6a) produces a different emission profile, with dominant features near 370 nm and 509 nm. The emission around 370 nm is attributed primarily to near-band-edge or host-related transitions in the IZO matrix, which may be influenced by erbium-induced modifications of the local electronic structure. The emission

near 509 nm is consistent with Er^{3+}-related transitions, such as $^4S_{3/2} \rightarrow {}^4I_{15/2}$, as reported in erbium-doped oxide systems. The comparatively broader spectral features observed under 300 nm excitation suggest the involvement of multiple radiative pathways, potentially including host-related recombination and defect-assisted luminescence processes, in addition to erbium-centered emission (Farid et al. 2016, Yamanaka et al. 2019). The dependence of photoluminescence intensity on Er sputtering power indicates that increasing erbium incorporation enhances the overall emission intensity (20 W < 30 W < 40 W), reflecting an increased population of optically active centers. However, this trend should be interpreted qualitatively, as higher dopant concentrations may also introduce defect-related non-radiative pathways that influence emission efficiency.

Colorimetric analysis based on CIE 1931 chromaticity diagrams (Figs. 6c–d) further illustrates the excitation-dependent emission behavior of the Er-doped IZO films. Under 380 nm excitation, the Er-doped samples cluster in the yellowish-white region of the chromaticity space (Mehta 2005) with the 20 W and 30 W samples approaching near-neutral white coordinates, while the 40 W sample shows a slight shift toward yellow (Djurišić et al. 2006).Under 300 nm excitation, all samples exhibit blue to blue–green emission, dominated by host-related transitions and modified by erbium incorporation. These results demonstrate the tunable optical response of Er-doped IZO thin films as a function of dopant concentration and excitation wavelength.

3.6 UV-vis spectroscopy analysis

Figure 7a-c indicates the UV-vis spectroscopy data, which reveals an obvious trend in the absorption as well as transmission properties with the augmentation of Er doping level. The transmission spectra indicates that all Er-doped films attain high transparency in the

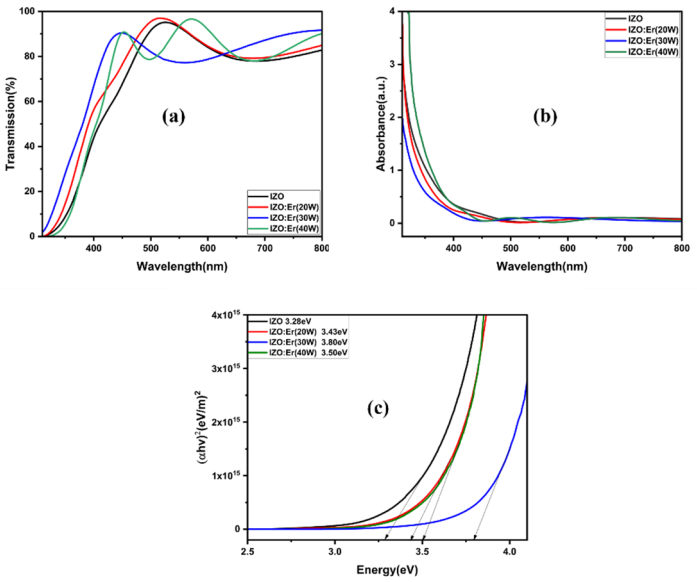


Fig. 7 **a** Transmission spectra of pure and Er doped IZO thin films, **b** Corresponding absorbance spectra, and **c** Tauc plots method for direct band gap materials used to estimate the optical bandgap of the doped IZO samples

visible region (>80%) (See Fig. 7a), which is essential for use in for photovoltaic transparent electrode applications (Aparna et al. 2023). Nonetheless, the transmission features are very slightly different: the 20 W Er-doped sample indicates the strongest transmission in the visible spectrum and the 40 W one possesses slightly lower transmission, which could be explained by carrier-induced absorption or structural scattering caused by Er-related defects (Sood et al. 2025, Silva Cardoso et al. 2020). As Er is further doped, the transmission cut-off edge shifts towards shorter wavelengths, as observed in the absorption measurements, which show an increase in the bandgap. As seen in the absorption spectra (See Fig. 7b), all the films have high UV absorption up to ~380 nm and quite low absorption in the visible part of the spectrum, which enables the conservation of the transparency properties of these films (Kim and Leem 2017). The absorption edge shifts consistently to higher energies (blue shift) as the concentration of Er increases, which is directly proportional to the widening bandgap. Such a blue shift means that Er doping is effective in creating shallow donor levels and increasing the concentration of free carriers, occupying the lower lying levels in the conduction band and compelling the optical transitions to take place at higher energies (Dixon et al. 2016).

The pristine IZO thin film exhibits an optical bandgap of 3.28 eV (See Fig. 7c). Upon Er doping, the bandgap shows a stepwise increase, reaching 3.43 eV at 20 W and 3.80 eV at 30 W of Er power, followed by a slight decrease to 3.50 eV at 40 W. This bandgap widening at moderate Er doping is a characteristic signature of the Burstein-Moss (BM) shift (WU et al. 2016), arising from the upward movement of the Fermi level into the conduction band as the carrier concentration increases due to Er^{3+} incorporation. In this regime, conduction-band states become filled, forcing optical transitions to occur at higher energies, thereby increasing the measured optical bandgap beyond the fundamental gap of IZO, consistent with the band-filling mechanism (Dixon et al. 2016). At higher doping power (40 W), the reduction from 3.80 to 3.50 eV indicates the onset of compensating mechanisms that begin to counteract the BM shift. Excessive Er incorporation can introduce defect complexes, lattice distortion, and increased ionized impurity scattering, all of which reduce the effective carrier concentration by trapping free electrons or localizing them. As a result, the net BM shift weakens, causing a partial narrowing of the apparent bandgap. Importantly, the system has not yet reached the carrier concentration threshold at which bandgap narrowing mechanisms (such as many-body band renormalization) fully dominate, but their initial influence becomes noticeable at this high doping level. Such a trend BM-driven widening at moderate doping and saturation or slight reduction at higher doping is widely observed in other rare-earth-doped transparent conducting oxides (Sood et al. 2025, Silva Cardoso et al. 2020).

3.7 Ellipsometry results

The refractive index, as illustrated in Fig. 8a, shows that the normal values around 2.1–1.8 in the visible-NIR range are characteristic of IZO, reported for transparent conducting oxides (Janicek et al. 2021, WU et al. 2016). When Er was doped, a systematic increase in the refractive index was observed at all wavelengths. The 20 W sample with Er doping has the maximum refractive index and 30 W and 40 W Er powers exhibit gradual reductions. This reaction may be explained by polarizability shifts caused by the introduction of Er^{3+} into the IZO lattice. The fact that the refractive index increases at moderate Er doping levels (through the increase in highly polarizable rare-earth ions) is attributed to the provision of

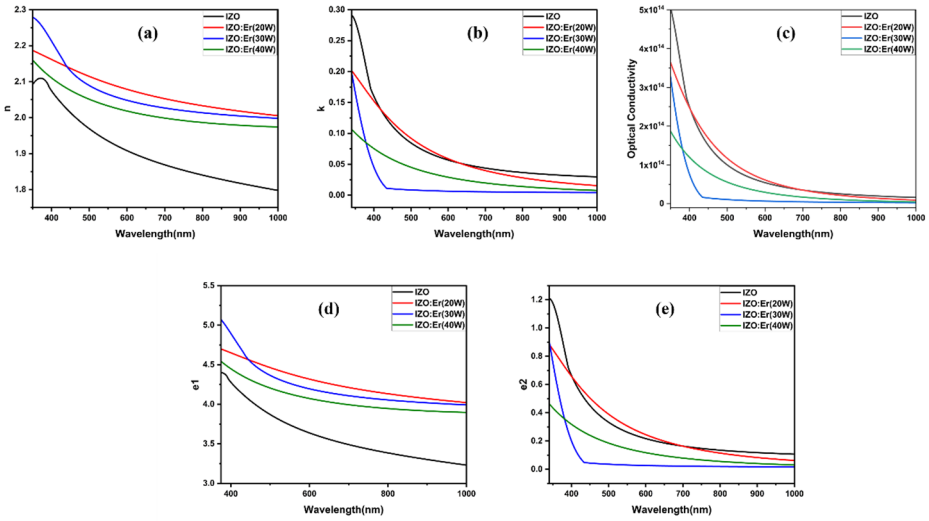


Fig. 8 Optical analysis of the synthesized thin films, showing (**a**) the variation in refractive index, **b** extinction coefficient profiles, **c** optical conductivity, **d** the real component, and **e** the imaginary component of the dielectric constant across different doping levels

more electronic transitions that alter the local dielectric environment (Singh and Kumar 2021). The decreasing trend towards higher Er concentrations could be caused by changes in the structure or appearance of Er-rich phases, which changes the total optical density in the material (Ahmed et al. 2023).

The extinction coefficient data in Fig. 8b shows the absorption behavior, which depends on the Er doping concentration. Pure IZO is highly transparent with a small absorption in the visible spectrum as is common in transparent conducting oxides (Afre et al. 2018). Er doping also has extra absorption features exhibited by the 20 W sample as having an increased absorption in the UV-visible region, and the other 30 W and 40 W samples displayed differences in their absorption profiles. This performance is explained by Er^{3+} intra 4f electronic transitions and charge processes in the states of Er and the IZO conduction band (Acosta-Silva et al. 2023). The trending extinction coefficient variation accompanies measurable variation in the bandgap, and it shows a dependence on the concentration of the process, which incorporates Er dopants into the optically active positions (Silva Cardoso et al. 2020, Al Mashary et al. 2024).

The optical conductivity (Fig. 8c) spectra agree with the metallic nature of the Er-doped IZO films and evident Drude-like behavior in the NIR region, which reflects the characteristics of a degenerate semiconductor (Ortega et al. 2019). The optical conductivity of pristine IZO is not very high, whereas Er doping contributes to the free-carrier response to a large extent. The frequency range of peaks centered at 3.0–4.0 eV of optical conductivity is associated with interband transitions, and these transitions are opened by Er doping. The gradual modifications of optical conductivity as a function of the concentration of Er points to variations in both the density of free carriers and free carrier mobility, as in the Drude model parameters obtained through ellipsometry analysis (WU et al. 2016, Krajewski et al. 2023).

The real part of the complex dielectric function (ε_1) (Fig. 8d) describes the ability of the material to store electric field energy and is related to the polarization response of the medium. Values ranging from approximately 3.0 to 5.5 are typical for oxide semiconductors and correlate directly with the refractive index through the relationship $\varepsilon_1 = n^2 - k^2$ (when $k << n$). The normal dispersion (decreasing value of ε_1 as the wavelength increases) is reflected in the frequency dependence of electronic polarization mechanisms. Systematic variation with the Er doping suggests that Er incorporation alters the local dielectric environment, perhaps due to alteration of the electronic density of states and polarization mechanisms. This parameter is basic to understand about the electromagnetic wave propagation in the material and design the optical interference structure (Cirilo-Lombardo 2015).

The imaginary part of the complex dielectric function (ε_2) (Fig. 8e) is a measure of the energy loss due to the absorption processes, and it relates to the extinction coefficient by the relationship $\varepsilon_2 = 2nk$. The relatively low values (≤ 1.0) confirm the low levels of optical losses in the visible spectrum, which is consistent with the transparent nature of the films. The wavelength dependence and the changes as a function of Er doping concentration give information about electronic transitions and absorption mechanisms. Higher values at shorter wavelengths indicate stronger interband absorption near the fundamental bandgap, while systematic changes with Er content suggest that erbium states introduce additional absorption channels without severely compromising transparency. This parameter is directly related to the absorption coefficient and is crucial for calculating optical losses in device applications (Cirilo-Lombardo 2015).

3.8 Work function

The experimentally determined values of the average work function (see Fig. 9a) indicate a typical trend, that is, pure IZO has a work function of 4.18 eV (Gao et al. 2015), as found in the literature on IZO thin films (Aydin et al. 2021). Upon doping with Er, the work function initially decreases to a minimum of 3.92 eV at 20 W sputtering power, followed by a slight

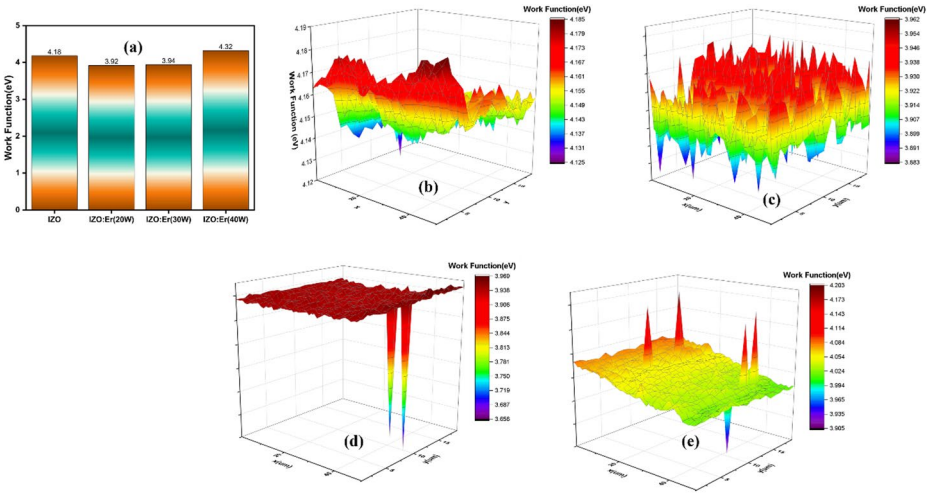


Fig. 9 **a** Average Work Function of thin films, 3D work function variation of (**b**) Pure **c** 20 W **d** 30 W **e** 40 W thin films

increment to 3.94 eV at 30 W, and finally a significant rise to 4.32 eV at 40 W. This non-monotonic trend reflects a complex interaction between dopant incorporation mechanisms and the electronic structure of the IZO matrix (Cai et al. 2022, Ndione et al. 2016).

At lower doping levels (20 W), Er^{3+} ions are substitutionally incorporated into the lattice, generating donor-type states that elevate the free carrier concentration. This shifts the Fermi level toward the conduction band, thereby decreasing the work function (Gao et al. 2015). This behavior aligns with the common trend observed in rare-earth doping of TCOs, where moderate doping is utilized to enhance conductivity via increased free electron availability (Sambasivam et al. 2020, Sood et al. 2025). The 3D work function (Fig. 9b-e) mapping for the 20 W sample confirms a uniform distribution, indicating the homogeneous character of Er incorporation throughout the film thickness. As the sputtering power is increased to 30 W, the work function remains relatively stable at 3.94 eV, indicating an optimal doping environment where Er atoms are controlled within substitutional sites without generating significant defect levels. The 3D mapping for this condition shows minor surface variations, yet the overall distribution remains uniform, signifying high film quality (Drewelow et al. 2019, Perumal et al. 2025). However, at the highest doping power (40 W), the work function rises considerably to 4.32 eV, a value exceedingly even that of pristine IZO. This increase is counterintuitive when considering bulk properties alone, as the 40 W sample exhibits the highest bulk carrier concentration $1.45 \times 10^{20} cm^{-3}$, which would typically suppress the work function via the Burstein-Moss effect. This anomaly suggests a decoupling of surface and bulk electronic properties, where surface-driven phenomena begin to dominate. Specifically, the degradation in Hall mobility at 40 W suggests increased ionized impurity scattering and defect density; these defects can create acceptor-like surface states that trap electrons, inducing upward band bending and increasing the effective surface potential barrier. Furthermore, the sharp rise in work function at 40 W can be attributed to surface chemical modifications. The high oxygen affinity of Er promotes the formation of Er–O bonds and potential partial surface segregation Er_2O_3, which possesses a higher intrinsic work function than the IZO host. This is supported by the possible formation of Er-enriched regions or clusters, which shift the surface chemistry and electronic structure (Baddorf 2023, Mirzai 2023). Additionally, the high-power sputtering process (40 W) imparts elevated kinetic energy to the emitted atoms, which may disrupt equilibrium growth conditions and introduce structural defects or interstitial Er atoms (Ji et al. 2025). These defects act as carrier traps, modifying the near-surface electronic structure and enhancing the surface dipole contribution, which counteracts the bulk Fermi-level shift. The 3D work function mapping at 40 W corroborates this, depicting strong inhomogeneity with random peaks and valleys that suggest phase separation and a random distribution of Er. Consequently, the observed trend is governed by a competition between the bulk Fermi-level shift dominant at moderate doping and surface dipole formation and defect-induced pinning, which dominate at high sputtering powers.

3.9 Electrical results

The electrical resistivity (depicted in Fig. 10a) exhibited a unique non-monotonic behavior with the concentration of Er doping. Pure IZO displays a resistivity of around 7×10^{-3} – Ω.cm which is as expected with undoped IZO films (Benouis et al. 2010, Morales-Masis et al. 2015). After doping Er at 20 W resistivity reduces greatly to approximately 2×10^{-3}

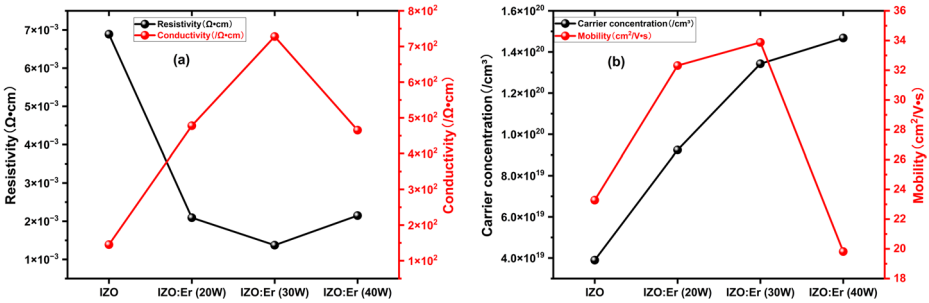


Fig. 10 **a** Comparative analysis of electrical resistivity and conductivity for IZO thin films modified with varying Er dopant concentrations (**b**) Variation in charge carrier mobility and density across different doping powers of Er in IZO

Ω.cm showing better electrical conductivity because the Er^{3+} was an effective donor (Sunde et al. 2014). Nevertheless, as the power of Er doping energy raises to 30 W, the resistivity drops to a minimum of around 1.5×10^{-3} Ω.cm which is the ideal state of doping. Even further increasing the Er power to 40 W results in further increased resistivity of approximately 2×10^{-3} Ω.cm implying more of these over-doping effects such that due to excessive incorporation of Er, electrical properties start to worsen (Ren et al. 2019, Dixon et al. 2016, Krajewski et al. 2023). Resistivity has an inverse relationship with conductivity (see Fig. 10a), as expected because of the reciprocal relationship between them ($\sigma = 1/\rho$) (Stadler 2012). Using undoped IZO, which has a conductivity of 1.4×10^2 $(\Omega.cm)^{-1}$, the Er-doped sample with a power of 20 W doubles the conductivity to approximately 4.8×10^2 $(\Omega.cm)^{-1}$ owing to the extra free carrier concentration of Er^{3+} donor states (Sunde et al. 2014, Shi et al. 2019). This was confirmed to have the optimum level of doping at 30 W Er because a maximum conductivity of approximately 7×10^2 $(\Omega.cm)^{-1}$ was attained. Conductivity also drops to approximately 4.8×10^2 $(\Omega.cm)^{-1}$ at 40 W Er doping, in agreement with the increase in resistivity and a representing the harmful impact of excess Er doping (Ren et al. 2019).

The carrier concentration (Fig. 10b) measurements showed the most dramatic change in the case of Er-doping. As shown in Pristine IZO, the carrier concentration is relatively low 4×10^{19} cm^{-3}, that is characteristic of lightly doped transparent conducting oxides (Morales-Masis et al. 2015, Stadler 2012). Er doping at 20 W brings substantial increase in the carrier concentration 9×10^{19} cm^{-3}, which is an indication that Er^{3+} forms an effective replacement of In^{3+} or occupies interstitial sites, giving extra free electrons (Ahmed et al. 2023). The carrier concentration is also increased significantly to 1.35×10^{20} cm^{-3} with er dopant at 30 W, which is the optimum point in dopant addition and the electrical characteristics (Krajewski et al. 2023, On et al. 2020). The value of the carrier concentration gains again to its peak or maximum value of about of 1.45×10^{20} cm^{-3}, at the 40 W Er doping, a phenomenon that indicates further incorporation of Er, despite the overall electrical performances starting to fall back (Cai et al. 2022). This behavior is characteristic of the classical trade-off in heavily doped TCOs of increased carrier density and decreased mobility attributed to increased scattering to ionized impurities and structural defects. The most complicated behavior is shown for the Hall mobility (Fig. 10b), that is the interdependence of different scattering processes in films that contain Er. The mobility of pristine IZO is approximately 23 $cm^2/V \cdot s$, which is within the acceptable range for amorphous or polycrystalline IZO films (Morales-Masis et al. 2015). When Er is doped at RT to 20 W, the mobility becomes approximately

Table 4 Optical–electrical parameters and Haacke figure of merit (Φ_{TC}) of pure and Er-doped IZO thin films, illustrating the trade-off between visible transparency and sheet resistance

Sample	T_{avg} (400–800 nm)	Resistivity, ρ (Ω·cm)	Thickness, t (cm)	Sheet Resistance, R_s (Ω/□)	FoM
IZO	0.80	6.89×10^{-3}	1.94×10^{-5}	3.55×10^{2}	3.02×10^{-4}
Er: IZO (20 W)	0.83	2.09×10^{-3}	2.02×10^{-5}	1.04×10^{2}	1.50×10^{-3}
Er: IZO (30 W)	0.84	1.37×10^{-3}	2.04×10^{-5}	6.74×10^{1}	2.60×10^{-3}
Er: IZO (40 W)	0.83	2.15×10^{-3}	2.47×10^{-5}	8.70×10^{1}	1.78×10^{-3}

32 $cm^2/V\cdot s$, which indicates a possible decrease in particular forms of defect scattering with moderate Er doping (On et al. 2020). The mobility attains a maximum of 34 $cm^2/V\cdot s$ at an Er doping of 30 W, which is the best film quality with minimal scattering owing to ionized impurities. Nevertheless, the mobility decreased dramatically to approximately 20 $cm^2/V\cdot s$ at 40 W of Er doping, even though the carrier concentration was the highest. Such degradation of mobility is explained by higher ionized impurity scattering caused by too many Er dopants, granular boundary scattering during structural transitions, or the development of Er-enrichment sub-phases causing increased scattering.

The transparent electrode performance of pure and Er-doped IZO thin films was evaluated using Haacke's figure of merit ($\Phi_{TC}=T^{10}/Rs$), which accounts for the trade-off between optical transparency and electrical conductivity. As summarized in Table 4, the FoM of undoped IZO is relatively low ($3.02\times10^{-3}\ \Omega^{-1}$), reflecting its high sheet resistance despite good optical transparency. Upon Er incorporation, Φ_TC increases significantly, reaching $1.50\times10^{-3}\ \Omega^{-3}$ for the 20 W Er-doped film. The highest FoM value of $2.60\times10^{-3}\ \Omega^{-1}$ is obtained for the 30 W Er-doped film, indicating an optimal balance between visible transparency and electrical transport. Further increase of Er sputtering power to 40 W results in a reduced FoM ($1.78\times10^{-3}\ \Omega^{-1}$), primarily due to increased sheet resistance associated with excessive dopant incorporation and defect-related scattering. These results demonstrate that moderate Er doping (30 W) yields the most favorable transparent electrode performance among the investigated samples.

4 Conclusion

The present study demonstrates that the structural, optical, and electronic properties of Er-doped indium zinc oxide thin films can be systematically tuned by controlling the erbium sputtering power during deposition. X-ray diffraction results show that moderate erbium incorporation (30 W) enhances crystallinity and reduces micro strain, whereas higher doping levels (40 W) introduce increased disorder and limit crystallite coherence. Optical characterization confirms that all Er–IZO films maintain high visible transparency (> 80%), while key optical parameters such as the bandgap, refractive index, and extinction coefficient can be effectively modulated through erbium concentration. The observed non-monotonic evolution of the work function reflects a balance between dopant activation, carrier trapping, and defect formation. Among the investigated conditions, the 30 W Er-doped film

exhibits an optimal combination of structural integrity, tunable optical response, and stable electronic behavior. Overall, these findings provide insight into the relationship between dopant-induced structural modifications and functional properties, and offer a materials-level foundation for the design of Er-doped IZO thin films as transparent conducting layers in optoelectronic and photovoltaic-relevant applications.

Acknowledgements The authors gratefully acknowledge the support received by Centre of Excellence in Solid State Physics at the University of the Punjab, Pakistan, and Department of Electronic & Computer Engineering, The Hong Kong University of Science and Technology Clear Water Bay, Kowloon, Hong Kong, SAR, China for providing all the facilities regarding experimentation and characterizations. The authors also acknowledge the support received from GIKI University for providing all the facilities regarding experimentation.

Author contributions Conceptualization, M.K. and Q.A.; methodology, M.K., Q.A., R.A. and N.A.; software, M.K. and M.B.A.; validation, Q.A. and M.B.A, G. A. N.; formal analysis, M. S, S.A. and S.N.; investigation, M.K. and R.A.; resources, R.A. and M.C.; data curation, S.A., and Q.A.; writing—original draft preparation, M.K. and Q.A.; writing—review and editing, S.N., R.A. and S.A.; supervision, M.C. and S.N.; project administration, S.A., S.N., and M.C. All authors have read and agreed to the published version of the manuscript.

Funding Open access funding provided by Hong Kong University of Science and Technology. Not applicable.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval Not applicable.

Consent to participate Not applicable.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Acosta-Silva, Y.J., et al.: Study of the effects of Er doping on the physical properties of CdSe thin films. Magnetochemistry. **9**(4), 107 (2023)

Afre, R.A., et al.: Transparent conducting oxide films for various applications: A review. Reviews Adv. Mater. Sci. **53**(1), 79–89 (2018)

Ágoston, P., et al.: Intrinsic n-Type behavior in transparent conducting oxides: format? a comparative hybrid-functional study of In 2 O 3, SnO 2, and ZnO. Phys. Rev. Lett. **103**(24), 245501 (2009)

Ahmed, M., et al.: Structural, Optical, Electric and Magnetic Characteristics of (In1 – xGdx) 2O3 Films for Optoelectronics. Materials. **16**(6), 2226 (2023)

Al Mashary, F.S., et al.: Effect of erbium-doping concentration on the electrical, structural and morphological properties of heterostructures based on TiO2 thin films. Bull. Mater. Sci. **47**(1), 42 (2024)

Ali, Q., et al.: Dopant-induced structural and electronic modifications in Ce–Er Co-doped IZO transparent conductive thin films toward advanced optoelectronics. Opt. Quant. Electron. **58**(1), 62 (2026)
Aparna, C., Shetty, P.K., Mahesha, M.: Investigation on structural, optical and electrical properties of Zn doped indium oxide thin film for gamma dosimetry. Mater. Chem. Phys. **302**, 127712 (2023)
Aydin, E., et al.: Ligand-bridged charge extraction and enhanced quantum efficiency enable efficient n–i–p perovskite/silicon tandem solar cells. Energy Environ. Sci. **14**(8), 4377–4390 (2021)
Baddorf, A.P.: Identifying the secondary electron cutoff in ultraviolet photoemission spectra for work function measurements of non-ideal surfaces. Sci. Rep. **13**(1), 13452 (2023)
Benouis, C., et al.: The effect of indium doping on structural, electrical conductivity, photoconductivity and density of states properties of ZnO films. J. Alloys Compd. **490**(1–2), 62–67 (2010)
Bouabdalli, E.M., et al.: Synthesis, structural, and optical behavior of erbium-doped silicophosphate glasses for photonics applications. Luminescence. **39**(6), e4802 (2024)
Cai, W., et al.: Performance enhancement of solution-processed InZnO thin-film transistors by Al doping and surface passivation. J. Semicond. **43**(3), 034102 (2022)
Christiansen, J., et al.: Strongly enhanced upconversion in trivalent erbium ions by tailored gold nanostructures: toward high-efficient silicon-based photovoltaics. Sol. Energy Mater. Sol. Cells. **208**, 110406 (2020)
Cirilo-Lombardo, D.J.: Semiconductor dielectric function, excitons and the Penn model. Phil. Mag. **95**(9), 1007–1015 (2015)
da Silva Cardoso, R., et al.: Structural, optical and photocatalytic properties of erbium (Er 3+) and yttrium (Y 3+) doped TiO 2 thin films with remarkable self-cleaning super-hydrophilic properties. RSC Adv. **10**(29), 17247–17254 (2020)
Dixon, S.C., et al.: n-Type doped transparent conducting binary oxides: an overview. J. Mater. Chem. C. **4**(29), 6946–6961 (2016)
Djurišić, A., et al.: Green, yellow, and orange defect emission from ZnO nanostructures: influence of excitation wavelength. Appl. Phys. Lett. (2006). https://doi.org/10.1063/1.2182096
Drewelow, G., et al.: Work function investigations of Al-doped ZnO for band-alignment in electronic and optoelectronic applications. Appl. Surf. Sci. **484**, 990–998 (2019)
Farid, S., et al.: Enhanced optical properties due to indium incorporation in zinc oxide nanowires. Appl. Phys. Lett. (2016). https://doi.org/10.1063/1.4939454
Gao, Y., et al.: The energy band tailored by Al incorporation in solution-processed IZO TFTs. RSC Adv. **5**(47), 37635–37639 (2015)
Ibiyemi, A., et al.: Synthesis and characterization of ZnO crystalline powder: effect on Alkaline pH. Afr. Sci. Rep. **4**, 225–225 (2025)
Janicek, P., et al.: Spectroscopic Ellipsometry Characterization of As-Deposited and Annealed Non-Stoichiometric Indium Zinc Tin Oxide Thin Film. Materials. **14**(3), 578 (2021)
Ji, F., et al.: The effects of sputtering powers on the structural, optical and electrical properties of β-Ga2O3 thin films prepared by magnetron sputtering. Phys. B: Condens. Matter **713**, 417322 (2025)
Juraić, K., et al.: Surface morphology of textured transparent conductive oxide thin film seen by various probes: visible light, X-rays, electron scattering and contact probe. Materials. **15**(14), 4814 (2022)
Kien, N., et al.: New insights on the optical properties and upconversion fluorescence of Er-doped CoAl 2 O 4 nanocrystals. RSC Adv. **14**(6), 3712–3722 (2024)
Kim, B., Leem, J.-Y.: Effects of doping concentration on the structural and optical properties of spin-coated in-doped ZnO thin films grown on thermally oxidized ZnO film/ZnO buffer layer/mica substrate. Korean J. Met. Mater. **55**(1), 67–71 (2017)
Koida, T., Ueno, Y., Shibata, H.: 2O3-based transparent conducting oxide films with high electron mobility fabricated at low process temperatures. Phys. status solidi (a). **215**(7), 1700506 (2018)
Krajewski, M., et al.: Structural, optical, and electrical properties of hafnium–aluminum–zinc-oxide films grown by atomic layer deposition for TCO applications. ACS omega. **8**(33), 30621–30629 (2023)
Kumi Barimah, E., et al.: Erbium-doped nanoparticle–polymer composite thin films for photonic applications: structural and optical properties. ACS omega. **5**(16), 9224–9232 (2020)
Kuwano, K., et al.: Effects on p-type buffer layers for semitransparent organic photovoltaics using indium zinc oxide transparent electrode. Jpn. J. Appl. Phys. **59**(SD), SDDD04 (2019)
Lee, D., Lee, A., Kim, H.-D.: IZO/ITO double-layered transparent conductive oxide for silicon heterojunction solar cells. IEEE Access. **10**, 77170–77175 (2022)
Lee, Y.-S., et al.: Effects of Deposition Power and Annealing Temperature on Indium Zinc Oxide (IZO) Film's Properties and Their Applications to the Source–Drain Electrodes of Amorphous Indium Gallium Zinc Oxide (a-IGZO) Thin-Film Transistors (TFTs). Nanomaterials. **15**(11), 780 (2025)
Li, J., et al.: Modulation of the TCO/MoO x front contact enables>21% high-efficiency dopant-free silicon solar cells. ACS Appl. Energy Mater. **6**(1), 285–294 (2022)

Li, B., et al.: Atomic-layer-deposition-free monolithic Perovskite/silicon tandem solar cell reaching 29.91% power conversion on industrial PERX/TOPCon-like silicon bottom cells. ACS Energy Lett. **9**(9), 4550–4556 (2024)
Liao, J., et al.: Green Upconversion Emission of Yb3+/Er3 + co-doped Gd2Sn2O7 Nanophosphors as Wide-range Temperature Sensor. J. Adv. Opt. Photonics. **1**(2), 95 (2018)
Makir, Z., et al.: Hafnium and erbium codoped ZnO thin films as anode material and spectral modificater for solar cells applications. (2021)
Mehta, A.: Map colors of a CIE plot and color temperature using an RGB color sensor. Strateg. Applicat. Eng. Maxim Integr. Prod. A **1026**, 1–11 (2005)
Mirzai, A.: Theoretical investigation of Rare-Earth doped optical materials: Optimization and Prediction. (2023)
Morales-Masis, M., et al.: Low-temperature high-mobility amorphous IZO for silicon heterojunction solar cells. IEEE J. Photovolt. **5**(5), 1340–1347 (2015)
Murugesan, M., et al.: Properties of Al-doped zinc oxide and In-doped zinc oxide bilayer transparent conducting oxides for solar cell applications. Mater. Lett. **222**, 50–53 (2018)
Ndione, P.F., et al.: Tuning the physical properties of amorphous In–Zn–Sn–O thin films using combinatorial sputtering. MRS Commun. **6**(4), 360–366 (2016)
On, N., et al.: Boosting carrier mobility and stability in indium–zinc–tin oxide thin-film transistors through controlled crystallization. Sci. Rep. **10**(1), 18868 (2020)
Ortega, J., et al.: Physical properties of reactive RF sputtered a-IZON thin films. Revista mexicana de física. **65**(2), 133–138 (2019)
Pan, H.-C., et al.: Influence of sputtering parameter on the optical and electrical properties of zinc-doped indium oxide thin films. J. Vacuum Sci. Technol. A. **23**(4), 1187–1191 (2005)
Park, S., et al.: The transparent conductive thin films of in2o3: zno for application of high efficiency perovskite/sitandem solarcell. Trans. Korean Inst. Electr. Eng. **70**(6), 887–892 (2021)
Perumal, R., Saravanan, L., Liu, J.-H.: Substrate and Doping Effects on the Growth Aspects of Zinc Oxide Thin Films Developed on a GaN Substrate by the Sputtering Technique. Processes. **13**(4), 1257 (2025)
Qiu, Q., et al.: ZnO-Doped In2O3 Front Transparent Contact Enables > 24.0% Silicon Heterojunction Solar Cells. Energy Technol. **11**(4), 2201310 (2023)
Quinn, D.J., Wardle, B., Spearing, S.M.: Residual stress and microstructure of as-deposited and annealed, sputtered yttria-stabilized zirconia thin films. J. Mater. Res. **23**(3), 609–618 (2008)
Ren, J., et al.: Effects of rare-earth erbium doping on the electrical performance of tin-oxide thin film transistors. J. Alloys Compd. **791**, 11–18 (2019)
Sambasivam, S., et al.: Effect of erbium on the structural, morphological, and optical properties of SnO2 thin films deposited by spray pyrolysis. Optik. **202**, 163596 (2020)
Seyhan, A., Kartal, E.: Optical, electrical and structural properties of ITO/IZO and IZO/ITO multilayer transparent conductive oxide films deposited via radiofrequency magnetron sputtering. Coatings. **13**(10), 1719 (2023)
Shi, Y.-L., et al.: High transmittance Er-doped ZnO thin films as electrodes for organic light-emitting diodes. Appl. Phys. Lett. (2019). https://doi.org/10.1063/1.5129065
Singh, P., Kumar, R.: Investigation of refractive index dispersion parameters of Er doped ZnO thin films by WDD model. Optik. **246**, 167829 (2021)
Sood, S., et al.: *Enhancing Optoelectronic Performance Through Rare-Earth-Doped ZnO: Insights and Applications*. in *Photonics*. MDPI (2025)
Stadler, A.: Transparent conducting oxides—an up-to-date overview. Materials. **5**(4), 661–683 (2012)
Sunde, T.O.L., et al.: Solid solubility of rare earth elements (Nd, Eu, Tb) in In 2 – x Sn x O 3–effect on electrical conductivity and optical properties. Dalton Trans. **43**(25), 9620–9632 (2014)
Taylor, M.P., et al.: The remarkable thermal stability of amorphous In-Zn-O transparent conductors. Adv. Funct. Mater. **18**(20), 3169–3178 (2008)
Tseng, C.-C., et al.: Effects of annealing on characteristics of Cu2ZnSnSe4/CH3NH3PbI3/ZnS/IZO nanostructures for enhanced photovoltaic solar cells. Nanomaterials. **10**(3), 521 (2020)
Wang, Y., et al.: Metal Nanowire-Induced Enhancement of Surface Emission in Luminescent Glass. Adv. Opt. Mater. **12**(36), 2401901 (2024)
WU, S.-W., LI, L.-Y., SAWADA, Y.: Spectroscopic ellipsometry study of the optoelectrical properties of In2O3: Sn–ZnO: Al thin films deposited through alternating sputtering. J. Ceram. Soc. Jpn. **124**(5), 528–531 (2016)
Yamanaka, M., et al.: Excitation of erbium-doped nanoparticles in 1550-nm wavelength region for deep tissue imaging with reduced degradation of spatial resolution. J. Biomed. Opt. **24**(7), 070501–070501 (2019)
Zahid, M.A., et al.: Influence of Al2O3/IZO double-layer antireflective coating on the front side of rear emitter silicon heterojunction solar cell. Vacuum. **200**, 110967 (2022)

Zeng, K., et al.: Investigation of mechanical properties of transparent conducting oxide thin films. Thin solid films. **443**(1–2), 60–65 (2003)

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.